📄 DOCUMENT SCANNER PROJECT REPORT

🟦 1. Introduction

The objective of this project is to design and implement a **document scanner using computer vision techniques**. The system takes an image of a document captured using a camera and transforms it into a clean, flat, and readable scanned version, similar to applications like CamScanner.

Traditional scanners require dedicated hardware, whereas this project demonstrates how software-based solutions can achieve similar results using image processing algorithms.

This project focuses on applying fundamental computer vision concepts such as:

- Edge detection
- Contour detection
- Perspective transformation
- Image enhancement

The implementation is done using Python, making use of libraries such as OpenCV and NumPy, along with a Streamlit interface for user interaction.

🟦 2. Methodology

The system follows a structured pipeline composed of several stages.

2.1 Image Preprocessing

The input image is first resized to improve computational efficiency while preserving the aspect ratio. Then, it is converted to grayscale to simplify processing, as color information is not necessary for detecting shapes.

A Gaussian blur is applied to reduce noise and smooth the image, which improves the quality of edge detection.

2.2 Edge Detection (Sobel Filter)

Edge detection is performed using the **Sobel operator**, which computes the gradient of the image intensity in both horizontal and vertical directions.

The Sobel filter highlights regions with high intensity variation, which correspond to edges. The gradients in the x and y directions are combined to produce the final edge map.

This step is crucial because it allows the system to identify the boundaries of objects, especially the document.

2.3 Contour Detection

Contours represent the boundaries of objects detected in the image. After obtaining the edge map, contours are extracted using OpenCV functions.

All detected contours are sorted based on their area, assuming that the document will be the largest object in the image.

▧ 3. Document Detection and Transformation

3.1 Detecting the Document

To identify the document, each contour is approximated to a simpler polygon using a curve approximation technique.

The contour that satisfies the following conditions is selected:

- Has the largest area
- Has exactly **four vertices**

A four-sided polygon corresponds to a rectangular shape, which is assumed to be the document.

3.2 Perspective Transformation

Once the four corners of the document are detected, a **perspective transformation** is applied.

This transformation maps the document from its original perspective (possibly tilted or skewed) into a top-down view, making it appear as if it were scanned by a flatbed scanner.

This step is essential for correcting distortions caused by the camera angle.

4. Image Enhancement

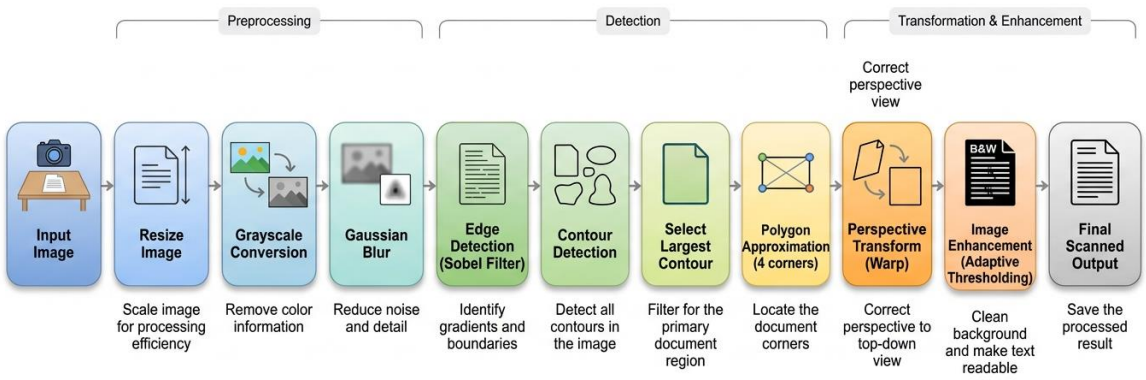
After obtaining the transformed image, enhancement techniques are applied to improve readability.

The image is converted to grayscale and processed using **adaptive thresholding**, which transforms it into a binary image (black and white).

This method adapts to local lighting conditions, effectively removing shadows and uneven illumination. As a result:

- The background becomes clean white
- The text becomes sharp and dark

This produces a professional “scanned document” appearance.



5. User Interface

To make the system user-friendly, a web interface was developed using **Streamlit**.

The interface allows users to:

- Upload an image
- Process the image using the scanning algorithm
- View results:
  - Original image
  - Warped (flattened) document
  - Enhanced scanned output
- Download the final scanned document

This makes the application interactive and easy to use without requiring technical knowledge.

6. Challenges Faced

During the development of this project, several challenges were encountered:

- Difficulty in detecting the correct contour when the background is complex
- Sensitivity to lighting conditions and shadows
- Ensuring correct ordering of the document corners for transformation
- Balancing performance and accuracy

These challenges were addressed through preprocessing techniques and parameter tuning.

7. Conclusion

This project demonstrates how classical computer vision techniques can be combined to create a practical and efficient document scanning application.

By using edge detection, contour analysis, and perspective transformation, it is possible to replicate the functionality of real-world scanning tools using only software.

The addition of a user interface further enhances usability, making the system accessible to non-technical users.

8. Future Improvements

Several improvements can be made to extend this project:

- Integration of OCR (Optical Character Recognition) to extract text
- Mobile application version
- Automatic brightness and contrast adjustment
- Support for multiple documents in one image